Quotient calibration: native source replay

aburan28/crypto · run-20261025T1845Z · October 25, 2026

The protocol and source preflight have been relocated from ghs-trapdoor PR 10. The original input hashes, kernel, training frame and 32 challenges are preserved. **Target calibration and inversion remain unexecuted.**

Figure 1: Solid edges: source checks executed. Dashed edges: target gates pending. Counts describe frozen input sets, not target successes.

Recorded checks

Check	Outcome
Source	F343 = F7[z]/(z³+z²+1), y² = x³+248x+42; G=(0,176), r=373, kernel 153.
Pilot	32 points × 64 sections; rank 8; selected indices 0–7.
Frame	G,2G,...,8G,11G; every eight of nine vectors have rank eight. All vectors and frame weights match.
Challenge freeze	32 inputs and P+G partners reproduced exactly; 99 sampling counters; no target images or challenge scalar labels.
Normalization	16 level-2 and 81 level-3 normal Weil-cocycle checks. Modulus irreducibility and source generator order checked.
Controls	Four Rust tests pass, including altered-manifest and singular-frame rejection. Strict Clippy passes.
Provenance	Four complete upstream input reports hash-verified; kept outside the evaluator.

Full normalized section-matrix SHA-256 matches the historical source:
`52f1938e51d9baca3e9d1753a90516611468ed1472c08203decbdd0a41ab87fe`

Execution and evidence

The command-line replay exited 0. Source freeze: `a0cd24e1c7f4381e9ce59cfce760eb5eb8cea3c5`

Rust 1.99.1, Cargo 1.99.1; Linux 6.18.55 x86_64; five available logical CPUs. Ordinary shared worker, no timing/isolation admission. No speed or memory-cost comparison is reported. Exact command, failed setup/lint attempts, provenance, root validation status and full interpretation are retained in REPORT.md and VALIDATION.md. The native-preflight.json receipt carries the measured counts.

The complete raw validation-log archive remains local and unpublished; automatic upload approval and AWS access are pending. Its hash and contents are recorded in ARCHIVE_PUBLICATION.json.

What this establishes

This is a separate native implementation check against historical Python/FLINT source outputs on one worker. The certified seed is fixed; its search is not repeated. No Python research process was executed. This finite source sample does not certify a global map identity, a target transfer or an attack gain.

The next step is to evaluate the nine target frame images, freeze M, then test all 32 target images exactly before inversion. Nine fitted projective frames alone supply no validation. Profile A retains the inherited anchor; Profile B still needs normalized target Weil functions and a global section certificate. The $F7^{18}$ coefficients also require compatible fields; the old $F7^{12}$ field cannot be reused. The complete acceptance gates remain in PROTOCOL.md.

Source: github.com/aburan28/ghs-trapdoor, commit b314cf87940cfb7199cd98261ba6d3910f275c90. Evidence: research/quotient_calibration_20261025/reports/run-20261025T1845Z/. No canonical cost graph or benchmark index changes.